

### (a) Cartesian Encoding

waypoints =  $(x_j, y_j, z_j)$



No intrinsic link  
to UAV motion  
Constraints via  
penalty functions

encoding  
→

### (b) Spherical Vector Encoding (SPSO)

segments =  $(\rho_j, \psi_j, \phi_j)$

UAV,  $(\psi_1, \phi_1)$   $(\rho_2, \psi_2, \phi_2)$



↓  
Constraints encoded  
in search bounds

↓  
Search in UAV's  
configuration space

$$x_{j+1} = x_j + \rho_j \sin \psi_j \cos \phi_j \quad y_{j+1} = y_j + \rho_j \sin \psi_j \sin \phi_j \quad z_{j+1} = z_j + \rho_j \cos \psi_j$$